

Bandit Algorithms

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The Multi-Armed Bandit (MAB) problem is a classic framework in decision theory, machine learning, and reinforcement learning that involves an agent sequentially choosing from several distinct options, or "arms," each providing a random reward from an unknown probability distribution. The core challenge is the exploration-exploitation trade-off: the agent must strategically explore by trying less-known arms to discover better rewards, while simultaneously exploiting the current best-known arm to maximize the cumulative reward over time. It serves as a foundational model for sequential decision-making under uncertainty, with applications ranging from clinical trials and advertising optimization (A/B testing) to resource allocation. We analyze a few well-known algorithms to address this problem.

1 Introduction and Problem Setup

We consider the classical stochastic *multi-armed bandit* (MAB) problem. There are k arms indexed by $a \in [A] = \{1, \dots, k\}$. Pulling arm a at time t yields a reward $X_t \sim P_a$, where the P_a are unknown distributions with finite means $\mu_a = \mathbb{E}_{P_a}[X]$. A policy (or strategy) π chooses arms sequentially, possibly stochastically, based on past observations. The interaction runs for a horizon n .

Definition 1. Let $\mathcal{A}$ be a finite set of arms. An (unstructured) **environment class** $\mathcal{E}$ is defined as

$$\mathcal{E} = \times_{a \in \mathcal{A}} \mathcal{M}_a$$

where $\mathcal{M}_a$ is a set of distributions of a for each $a \in \mathcal{A}$. An environment is an element in this class.

Remark 1. Environment classes can also be structured. Here, information of some arm can be obtained from another arm instead of the arms being completely independent of each other, as in the case of unstructured classes.

$\mathcal{H}_t = \{(a_1, x_1, \dots, a_t, x_t) : a_i \in \mathcal{A}, x_i \in \mathbb{R}\}$ is the set of all **histories**. Let n be a finite **horizon**, which denotes the total number of pulls/rounds so far.

1.1 Canonical Bandit Model

For each $t \in [n]$, $\Omega_t = ([k] \times \mathbb{R})^t$, and $\mathcal{F}_t = \mathcal{B}(\Omega_t)$

$$A_t : \Omega_n \rightarrow [k], A_t(a_1, x_1, \dots, a_n, x_n) = a_t$$

$$X_t : \Omega_n \rightarrow \mathbb{R}, X_t(a_1, x_1, \dots, a_n, x_n) = x_t$$

Definition 2. A **policy** π is a sequence $(\pi_t)_{t=1}^n$ where, π_t is a probability kernel from $(\Omega_t, \mathcal{F}_t)$ to $([k], 2^{[k]})$.

Remark 2. Let $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{G})$ be measure spaces. A **probability kernel** from $(\mathcal{X}, \mathcal{F})$ to $(\mathcal{Y}, \mathcal{G})$ is a function $\mathcal{K} : \mathcal{X} \times \mathcal{G} \rightarrow [0, 1]$ such that:

1. $\mathcal{K}(x, \cdot)$ is a measure for each $x \in \mathcal{X}$
2. $\mathcal{K}(\cdot, A)$ is $\mathcal{F}$ -measurable for each $A \in \mathcal{G}$

In essence, policy is a "strategy" which, for a sequence of decisions and rewards (which we defined as history), gives a distribution over the set of arms. Now, all the algorithms we look at are deterministic, which means that $\pi(a_t | a_1, x_1, \dots, a_{t-1}, x_{t-1}) = 1$ and $\pi(a_{t'} | a_1, x_1, \dots, a_{t-1}, x_{t-1}) = 0$ for $t' \neq t$, i.e. our strategy tells us exactly which arm to pick next. When this is not the case the algorithm becomes randomized (which we have not covered in this project).

Assume that the space of rewards for each arm is finite (and discrete) and the probability of the reward of arm a_t being x_t is $p_{a_t}(x_t)$. Then define,

$$p_{\nu\pi}(a_1, x_1, \dots, a_n, x_n) = \prod_{i=1}^n \pi(a_i | a_1, x_1, \dots, a_{i-1}, x_{i-1}) p_{a_i}(x_i)$$

Remark 3. $p_{\nu\pi}$ is a distribution over $([k] \times \mathbb{R})^n$ which tells us the probability of the occurrence of a particular history.

Remark 4. For rewards that aren't finite or even discrete, Radon-Nikodym can be used to show the existence of a probability distribution, which on integration, gives the distribution function P_a for each a .

1.2 Random Table Model

Let $\Omega = \mathbb{R}^{nk}$ and $\mathcal{F} = \mathcal{B}(\Omega)$.

Define random variables $(X_{si})_{s \in [n], i \in [k]}$ over $(\Omega, \mathcal{F})$ such that law of X_{ti} is $\mathcal{P}_i$.

Then let $X_t = X_{tA_t}$, where the actions A_t are $\mathcal{F}_{t-1}$ -measurable with $\mathcal{F}_{t-1} = \sigma(A_1, X_1, \dots, A_{t-1}, X_{t-1})$, where $\sigma(A_1, X_1, \dots, A_{t-1}, X_{t-1})$ is the smallest sigma algebra that makes each of $A_1, X_1, \dots, A_{t-1}, X_{t-1}$ a random variable.

1.3 Objective: Cumulative Regret

Let $\mu^* = \max_a \mu_a$ and define the gap $\Delta_a = \mu^* - \mu_a \geq 0$. If $T_a(n)$ denotes the number of times arm a is pulled in the first n rounds, the *expected regret* of policy π in environment $\nu = (P_1, \dots, P_k)$ is

$$R_n(\pi, \nu) := n\mu^* - \mathbb{E}\left[\sum_{t=1}^n X_t\right] = \sum_{a=1}^k \Delta_a \mathbb{E}[T_a(n)]. \quad (1)$$

The second equality can be proven as follows,

$$\begin{aligned} R_n &= n\mu^* - \mathbb{E}[S_n] = n\mu^* - \mathbb{E}\left[\sum_{t=1}^n X_t\right] = \mathbb{E}\left[\sum_{t=1}^n (\mu^* - X_t)\right] \\ &= \mathbb{E}\left[\sum_{a \in \mathcal{A}} \sum_{t=1}^n (\mu^* - X_t) \mathbb{I}_{a_t=a}\right] \\ &= \sum_{a \in \mathcal{A}} \sum_{t=1}^n \mathbb{E}[(\mu^* - X_t) \mathbb{I}_{a_t=a}] \end{aligned} \quad (2)$$

Now,

$$\begin{aligned} \mathbb{E}[(\mu^* - X_t) \mathbb{I}_{a_t=a}] &= \sum_{a \in \mathcal{A}} \mathbb{E}[(\mu^* - X_t) \mathbb{I}_{a_t=a} | a] \\ &= \mathbb{E}[(\mu^* - X_t) \mathbb{I}_{a_t=a} | a_t] \\ &= (\mu^* - \mu_{a_t}) \mathbb{E}[\mathbb{I}_{a_t=a}] = \Delta_{a_t} \mathbb{E}[\mathbb{I}_{a_t=a}] \end{aligned} \quad (3)$$

The second equality follows from

$$\mathbb{E}[(\mu^* - X_t) \mathbb{I}_{a_t=a} | s] = 0 \forall s \neq a$$

Thus, we get from (2), (3), and definition of $T_a(n)$,

$$R_n = \sum_{a \in \mathcal{A}} \Delta_a \mathbb{E}[T_a(n)]$$

Thus bounding regret reduces to controlling expected pulls of suboptimal arms.

2 Sub-Gaussian Random Variables

A key technical assumption used throughout is sub-gaussianity.

Definition 3. A random variable X is σ -**subgaussian** if for all $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{\lambda(X - \mathbb{E}X)}] \leq e^{\lambda^2 \sigma^2 / 2}$.

Theorem 1. If X is σ -subgaussian, then for any $\epsilon \geq 0$,

$$\mathbb{P}(X \geq \epsilon) \leq e^{-\epsilon^2 / 2\sigma^2}$$

Proof. Let $\lambda > 0$. Then,

$$\begin{aligned} \mathbb{P}(X \geq \epsilon) &= \mathbb{P}(e^{\lambda X} \geq e^{\lambda \epsilon}) \\ &\leq \mathbb{E}(e^{\lambda X}) e^{-\lambda \epsilon} \\ &\leq e^{\lambda^2 \sigma^2 / 2} e^{-\lambda \epsilon} \end{aligned} \quad (4)$$

The first inequality by Markov's inequality and the second by definition of σ -subgaussian. Choosing $\lambda = \epsilon / \sigma^2$ completes the proof. $\square$

An equivalent from of these bounds is

$$\mathbb{P}(X \geq \sqrt{2\sigma^2 \log(1/\delta)}) \leq \delta$$

On applying the union bound,

$$\mathbb{P}(|X| \geq \sqrt{2\sigma^2 \log(2/\delta)}) \leq \delta$$

Remark 5. Let X_1 and X_2 are σ_1 and σ_2 subgaussian and independent respectively. Then,

1. if $\mathbb{E}[X_1] = 0$ then $\text{Var}(X_1) \leq \sigma_1^2$
2. cX_1 is $|c|\sigma_1$ subgaussian for all $c \in \mathbb{R}$
3. $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$ subgaussian

From this we obtain Hoeffding-style tail bounds: for $\epsilon > 0$,

$$\mathbb{P}(\hat{\mu} \leq \mu + \epsilon) \leq e^{-\frac{n\epsilon^3}{2\sigma^2}}$$

$$\mathbb{P}(\hat{\mu} \geq \mu - \epsilon) \leq e^{-\frac{n\epsilon^3}{2\sigma^2}}$$

where $\hat{\mu} = \frac{1}{n} \sum_{t=1}^n X_t$.

Remark 6. Subgaussianity is satisfied by bounded rewards and many light-tailed distributions; it greatly simplifies confidence-bound constructions used by UCB-type algorithms.

3 Explore-Then-Commit Algorithm

Explore-then-Commit is one of the simplest algorithms and is described below.

Algorithm 1 ETC Algorithm

- 1: Input $\mathbf{m}$
- 2: For each round, t , choose;

$$A_t = \begin{cases} (t \bmod k) + 1, & \text{if } t \leq mk \\ \text{argmax}_i \hat{\mu}_i(mk), & \text{otherwise} \end{cases}$$

Here, $\hat{\mu}_i(mk)$ is the empirical mean of the reward from arm i over t time steps. To summarize, we first pull each arm m times, this is the exploration phase. Then, we exploit the arm that has the highest empirical mean.

We analyze the Regret of this policy in the environment $\mathcal{E}_{SG}^k(1)$, i.e k -armed bandits, with reward distribution being 1-Subgaussian.

Theorem 2. For horizon n , the regret is bounded above by,

$$R_n \leq m \sum_{i=1}^n \Delta_i + (n - mk) \sum_{i=1}^n \Delta_i \exp\left(\frac{-m\Delta_i^2}{4}\right) \quad (5)$$

for $1 \leq m \leq n/k$, where $\Delta_i = \mu^* - \mu_i$, is the suboptimality gap.

Proof. By Theorem 1 Regret can be decomposed as follows;

$$R_n = \sum_{i=1}^n \Delta_i \mathbb{E}[T_i(n)] \quad (6)$$

Without loss of generality, assume arm 1 is the optimal arm, thus, $\Delta_i = \mu_1 - \mu_i$

$$\begin{aligned} \mathbb{E}[T_i(n)] &= m + (n - mk) \mathbb{P}(A_{mk+1} = i) \\ &\leq m + (n - mk) \mathbb{P}(\hat{\mu}_i(mk) \geq \hat{\mu}_1(mk)) \end{aligned}$$

$$\mathbb{P}(\hat{\mu}_i(mk) \geq \hat{\mu}_1(mk)) = \mathbb{P}([\hat{\mu}_i(mk) - \mu_i] - [\hat{\mu}_1(mk) - \mu_1] \geq (\mu_i - \mu_1))$$

Note that $\hat{\mu}_i(mk)$ is a $\sqrt{(1/m)}$ -Subgaussian random variable. Then, $[\hat{\mu}_i(mk) - \mu_i] - [\hat{\mu}_1(mk) - \mu_1]$ is $\sqrt{(2/m)}$ -Subgaussian. So by the Hoeffding Inequality,

$$\mathbb{P}(\hat{\mu}_i(mk) \geq \hat{\mu}_1(mk)) = \mathbb{P}([\hat{\mu}_i(mk) - \mu_i] - [\hat{\mu}_1(mk) - \mu_1] \geq \Delta_i) \leq \exp\left(\frac{-m\Delta_i^2}{4}\right)$$

Substituting these, we get the required upper bound. $\square$

Remark 7. Note that the two terms of the upper bound, indicate the trade-off between the exploration and exploitation phases of the algorithm, balanced by the appropriate choice of m . An arbitrary choice of m cannot guarantee sublinear regret.

Consider the two-armed case as a toy example. Let $\Delta_1 = 0$ and $\Delta_2 = \Delta$.

$$R_n \leq m\Delta + (n - 2m)\Delta \exp\left(\frac{-m\Delta^2}{4}\right) \leq m\Delta + n\Delta \exp\left(\frac{-m\Delta^2}{4}\right) \quad (7)$$

Minimize the upper bound, by differentiating with respect to m and set to 0.

$$\begin{aligned} \Delta - n\frac{\Delta^3}{4} \exp\left(\frac{-m\Delta^2}{4}\right) &= 0 \\ \implies m &= \frac{4}{\Delta^2} \log\left(\frac{n\Delta^2}{4}\right) \\ \implies m &= \max\left\{1, \left\lceil \frac{4}{\Delta^2} \log\left(\frac{n\Delta^2}{4}\right) \right\rceil\right\} \end{aligned}$$

The last implication follows from the simple fact that $m \geq 1$ always. Next we substitute this choice of m back into (3) for our bound on Regret. This is done in two cases,

- Case 1 : $\frac{n\Delta^2}{4} \leq 1 \Rightarrow m = 1$

$$R_n \leq \Delta + n\Delta \exp\left(\frac{-\Delta^2}{4}\right) \leq \Delta + \frac{4}{\Delta}$$

The second inequality holds because $n \leq 4/\Delta^2$ and $e^{-\Delta^2/4} < 1$. But the above bound is trivial because in this case, $n\Delta \leq \Delta + \frac{4}{\Delta}$, which is trivially an upper bound on regret. Further, $\Delta \leq 2/\sqrt{n}$ implies that $R_n \leq 2\sqrt{n}$, thus sublinear in n .

- Case 2 : $\frac{n\Delta^2}{4} > 1 \Rightarrow m = \left\lceil \frac{4}{\Delta^2} \log\left(\frac{n\Delta^2}{4}\right) \right\rceil$

$$\text{Then } \frac{4}{\Delta^2} \log\left(\frac{n\Delta^2}{4}\right) \leq m \leq \frac{4}{\Delta^2} \log\left(\frac{n\Delta^2}{4}\right) + 1$$

$$\begin{aligned} m\Delta &\leq \Delta + \frac{4}{\Delta} \log\left(\frac{n\Delta^2}{4}\right) \\ n\Delta \exp\left(\frac{-m\Delta^2}{4}\right) &\leq n\Delta \frac{4}{n\Delta^2} = \frac{4}{\Delta} \\ \implies R_n &\leq \Delta + \frac{4}{\Delta} \left(1 + \log\left(\frac{n\Delta^2}{4}\right)\right) \end{aligned}$$

To show that R_n is sublinear in this case as well, define $x := \frac{\Delta\sqrt{n}}{2}$.

$$\begin{aligned} R_n &\leq \Delta + 4 \left(\frac{\sqrt{n}}{2x}\right) (1 + \log(x^2)) \\ &= \Delta + \sqrt{n} \cdot \frac{2(1 + 2\log x)}{2} \\ &\leq \Delta + \sqrt{n} \cdot \frac{4}{\sqrt{e}} \end{aligned}$$

Although, we have shown that ETC may guarantee nearly optimal regret, i.e. $R_n \leq \Delta + C\sqrt{n}$, there are a few glaring problems. Firstly, the regret is not as easily tractable in the multi-armed case, as it was above. Secondly, to find the optimal m , we require the knowledge of Δ and n . The latter is sometimes known, but it is unreasonable to assume that we know the former quantity. The requirement of Δ and n is fixed by making m data dependent rather than on these two quantities. But this also motivates us to look for better algorithms.

4 Upper Confidence Bound Algorithm

This is an **index algorithm**, which means that at each time step, an **index** is calculated for each arm, and the arm that maximizes it is chosen. Here the index is the upper bound of a confidence interval around the empirical mean. For arm i , the δ -UCB at time t is defined as

$$UCB_i(t, \delta) = \begin{cases} \infty & , \text{if } T_i(t) = 0 \\ \hat{\mu}_i(t) + \sqrt{\frac{2 \log(1/\delta)}{T_i(t-1)}} & , \text{otherwise} \end{cases}$$

The index is defined this way because we know that, for a sequence of *iid* 1-Subgaussian random variables with mean μ and $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_t$:

$$\mathbb{P} \left(\mu \geq \hat{\mu} + \sqrt{\frac{2 \log(1/\delta)}{n}} \right) \leq \delta \quad \forall \delta \in (0, 1)$$

The fundamental idea behind this algorithm is, we assume that the environment behaves as nicely as plausible. This means we fix a confidence level $1 - \delta$ and compute the highest possible reward for each arm, with probability at least $1 - \delta$. Then we choose the arm for which this is maximized.

Algorithm 2 UCB(δ) Algorithm

- 1: Input k and δ
 - 2: **for** $t \in 1, \dots, n$ **do**
 - 3: Choose $A_t = \arg \max_i UCB_i(t-1, \delta)$
 - 4: Observe the reward X_t and update UCB's
 - 5: **end for**
-

Ideally, if an algorithm chooses arm a , it should have done so because, it is a promising arm, i.e. $\hat{\mu}_a$ is high, or because the arm was not explored well enough, that means it was pulled too few times. In this case $T_a(t)$ is low, so the second term, the confidence width, is higher. This is the intuition behind the UCB algorithm. Choosing the arm with the largest upper confidence bound leads to a situation where arms are only chosen if their true mean could reasonably be larger than those of arms that have been played often.

Now, it remains to choose a reasonable confidence level δ . If at any time, UCB of the optimal arm drops below its true mean, the algorithms might stop pulling that arm, and it could suffer linear regret. To avoid this, we can take $\delta = o(1/n)$, say $1/n^2$. Again we analyze Regret for this choice of δ in the environment $\mathcal{E}_{SG}^k(1)$.

First we motivate the proof of the following theorem. As usual WLOG, assume arm 1 is optimal. Suppose after the round robin, i.e. after each arm is chosen at least once, at some time t , the arm i is chosen. This means that $UCB_i(t-1, \delta) > UCB_1(t-1, \delta)$ which can happen if at least one of the following events happen:

1. Suboptimal arm i was determined to be possibly optimal, i.e. $UCB_i(t-1, \delta) > \mu_1$
2. Optimal arm 1 was determined to be suboptimal, i.e., $UCB_1(t-1, \delta) < \mu_1$

Event 2 does not occur with high probability by construction of the confidence interval. Event 1 can only occur a finite number of times, because, as the suboptimal arm is pulled $\hat{\mu}_i \rightarrow \mu_i < \mu_1 < UCB_1(t, \delta)$, that is, it learns that i is indeed suboptimal. Now we formalize this idea.

Theorem 3. For any horizon n , if $\delta = 1/n^2$ the UCB(δ) algorithm suffers regret bounded above by

$$R_n \leq 3 \sum_{i=1}^k \Delta_i + \sum_{i: \Delta_i > 0} \frac{16 \log n}{\Delta_i} \quad (8)$$

Proof. By regret decomposition, $R_n = \sum_{i=1}^k \Delta_i \mathbb{E}[T_i(n)]$. We bound $\mathbb{E}[T_i(n)]$. For an arm i , define the "good event" G_i as follows;

$$G_i := \left\{ \mu_1 < \min_{t \in [n]} UCB_1(t, \delta) \right\} \cap \left\{ \hat{\mu}_{i, u_i} + \sqrt{\frac{2 \log(1/\delta)}{u_i}} < \mu_1 \right\}$$

where $u_i \in \mathbb{N}$ will be chosen later so that; (a) If G_i occurs, $T_i(n) \leq u_i$ and (b) G_i^c occurs with low probability. Suppose G_i did occur but $T_i(n) > u_i$. Then for some t , $T_i(t-1) = u_i$ and $A_t = i$. Then,

$$UCB_i(t-1, \delta) = \hat{\mu}_i(t-1) + \sqrt{\frac{2 \log(1/\delta)}{T_i(t-1)}} = \hat{\mu}_{i, u_i} + \sqrt{\frac{2 \log(1/\delta)}{u_i}}$$

But this is less than μ_1 and μ_1 is less than index of 1, because G_i occurred. Which means $A_t \neq i$, a contradiction. Now to bound $\mathbb{P}(G_i^c)$ apply the Union bound,

$$\begin{aligned}\mathbb{P}(G_i^c) &\leq \mathbb{P}\left(\mu_1 \geq \min_{t \in [n]} \text{UCB}_1(t, \delta)\right) + \mathbb{P}\left(\hat{\mu}_{i, u_a} + \sqrt{\frac{2 \log(1/\delta)}{u_a}} \geq \mu_1\right) \\ \mathbb{P}\left(\mu_1 \geq \min_{t \in [n]} \text{UCB}_1(t, \delta)\right) &\leq \sum_{s=1}^n \mathbb{P}\left(\mu_1 \geq \hat{\mu}_{1, s} + \sqrt{\frac{2 \log(1/\delta)}{s}}\right) \leq n\delta\end{aligned}$$

Next, to bound the second term, let us take u_i large enough so that

$$(1-c)\Delta_i \geq \sqrt{\frac{2 \log(1/\delta)}{u_i}} \implies \Delta_i - \sqrt{\frac{2 \log(1/\delta)}{u_i}} \geq c\Delta_i \quad (9)$$

for $c \in (0, 1)$, to be fixed later. Then, subtract μ_i from both sides of the inequality in the second term

$$\begin{aligned}\mathbb{P}\left(\hat{\mu}_{i, u_i} + \sqrt{\frac{2 \log(1/\delta)}{u_i}}\right) &= \mathbb{P}\left(\hat{\mu}_{i, u_i} - \mu_i \geq \Delta_i - \sqrt{\frac{2 \log(1/\delta)}{u_i}}\right) \\ &\leq \mathbb{P}(\hat{\mu}_{i, u_i} - \mu_i \geq c\Delta_i) \leq \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right)\end{aligned}$$

Finally, split the expectation as follows and apply the fact that $T_I(n) \leq u_i$ if G occurs.

$$\begin{aligned}\mathbb{E}[T_i(n)] &= \mathbb{E}[\mathbb{I}_{G_i} T_i(n)] + \mathbb{E}[\mathbb{I}_{G_i^c} T_i(n)] \leq u_i + n\mathbb{P}(G_i^c) \\ &\leq u_i + n\left(n\delta + \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right)\right)\end{aligned}$$

Naturally choose u_i to be the smallest integer such that condition 9 holds

$$u_i = \left\lceil \frac{2 \log(1/\delta)}{(1-c)^2 \Delta_i^2} \right\rceil \geq \frac{2 \log(1/\delta)}{(1-c)^2 \Delta_i^2}$$

Substitute the above inequality and $\delta = 1/n^2$ to obtain

$$\mathbb{E}[T_i(n)] \leq \left\lceil \frac{2 \log(1/\delta)}{(1-c)^2 \Delta_i^2} \right\rceil + 1 + n^{1-2c^2/(1-c)^2} \leq 3 + \frac{16 \log(n)}{\Delta_i^2}$$

The last inequality follows by choosing $c = 1/2$. Substitute this back into the regret decomposition equation to obtain the exact result. $\square$

Again, this bound is dependent on $1/\Delta_i$ terms which can blow up if the suboptimality gap is quite small for any arm. Thus, we prove the following **gap-free** bound on the regret of UCB.

Theorem 4. For any $\nu \in \mathcal{E}_{SG}^k(1)$, if $\delta = 1/n^2$, the regret of UCB is bounded by,

$$R_n \leq 8\sqrt{kn \log n} + 3 \sum_{i=1}^k \Delta_i$$

Proof. Fix $\Delta > 0$, whose value will be decided later. By the regret decomposition lemma and the above theorem, we get

$$\begin{aligned}R_n &= \sum_{i=1}^k \Delta_i \mathbb{E}[T_i(n)] = \sum_{i: \Delta_i < \Delta} \Delta_i \mathbb{E}[T_i(n)] + \sum_{i: \Delta_i \geq \Delta} \Delta_i \mathbb{E}[T_i(n)] \\ &\leq n\Delta + \frac{16k \log(n)}{\Delta} + 3 \sum_{i=1}^k \Delta_i \\ &\leq 8\sqrt{kn \log n} + 3 \sum_{i=1}^k \Delta_i\end{aligned}$$

We choose $\Delta = \sqrt{(16k \log n)/n}$ so that the last inequality holds. $\square$

5 Asymptotically Optimal UCB

This version of UCB still has the same driving principle. It only differs in the confidence level of the UCB, which varies with time, as opposed to the vanilla UCB where the confidence level is agnostic of the current timestep. Here, as t increases, we want to be increasingly sure that we are pulling the optimal arm.

Algorithm 3 Asymptotically Optimal UCB

- 1: Input k
- 2: Choose each arm once
- 3: Subsequently choose

$$A_t = \arg \max_i \left(\hat{\mu}_i(t-1) + \sqrt{\frac{2 \log f(t)}{T_i(t-1)}} \right) \quad \text{where } f(t) = 1 + t \log^2 t$$

The upper bound on regret for this algorithm is more complicated than that of the plain UCB, and the dominant terms are of the same order of growth. However, the improvement is in the leading constant which determines the asymptotic behaviour.

Theorem 5. *For any $\nu \in \mathcal{E}_{SG}^k(1)$, the regret of the Asymptotically Optimal UCB is bounded by*

$$R_n \leq \sum_{i: \Delta_i > 0} \inf_{\epsilon \in (0, \Delta_i)} \Delta_i \left(1 + \frac{5}{\epsilon^2} + \frac{2 \left(\log f(n) + \sqrt{\pi \log f(n)} + 1 \right)}{(\Delta_i - \epsilon)^2} \right) \quad (10)$$

Furthermore,

$$\limsup_{n \rightarrow \infty} \frac{R_n}{\log(n)} \leq \sum_{i: \Delta_i > 0} \frac{2}{\Delta_i} \quad (11)$$

Before we prove this theorem, observe that choosing $\epsilon = \Delta_i/2$ gives

$$\begin{aligned} R_n &\leq \sum_{i: \Delta_i > 0} \left(\Delta_i + \frac{1}{\Delta_i} \left(8 \log f(n) + 8 \sqrt{\pi \log f(n)} + 28 \right) \right) \\ \implies R_n &\leq C \sum_{i: \Delta_i} \left(\Delta_i + \frac{\log(n)}{\Delta_i} \right) \text{ for a universal constant } C > 0 \\ \implies R_n &\leq 2 \sqrt{C k n \log n} + C \sum_{i=1}^k \Delta_i \end{aligned}$$

The last implication follows similar to Theorem 4. This shows that this improved UCB is indeed of the same order as the vanilla UCB. Before we get into the proof, we first prove a necessary lemma.

Lemma 1. *Let $X_1, X_2, \dots, X_n$ be a sequence of iid 1-subgaussian random variables, $\hat{\mu}_t = \frac{1}{t} \sum_{i=1}^t X_i$, $\epsilon > 0, a > 0$ and*

$$\kappa = \sum_{t=1}^n \mathbb{I} \left\{ \hat{\mu}_t + \sqrt{\frac{2a}{t}} \geq \epsilon \right\}, \quad \kappa' = \frac{2a}{\epsilon^2} + \sum_{t=\lceil \frac{2a}{\epsilon^2} \rceil}^n \mathbb{I} \left\{ \hat{\mu}_t + \sqrt{\frac{2a}{t}} \geq \epsilon \right\} \quad (12)$$

Then $\mathbb{E}[\kappa] \leq \mathbb{E}[\kappa'] \leq 1 + \frac{2}{\epsilon^2}(a + \sqrt{\pi a} + 1)$.

Proof. The first inequality follows because $\hat{\mu}_t + \sqrt{2a/t}$ cannot be smaller than ϵ until $t \geq 2a/\epsilon^3$, since $\mathbb{E}[\hat{\mu}_t] = 0$.

$$\begin{aligned} \mathbb{E}[\kappa'] &= \frac{2a}{\epsilon^2} + \sum_{t=\lceil \frac{2a}{\epsilon^2} \rceil}^n \mathbb{P} \left(\hat{\mu}_t + \sqrt{\frac{2a}{t}} \geq \epsilon \right) \\ &\leq \frac{2a}{\epsilon^2} + \sum_{t=\lceil \frac{2a}{\epsilon^2} \rceil}^n \exp \left(-\frac{t(\epsilon - \sqrt{2a/t})^2}{2} \right) \\ &\leq 1 + \frac{2a}{\epsilon^2} + \int_{\frac{2a}{\epsilon^2}}^{\infty} \exp \left(-\frac{t(\epsilon - \sqrt{2a/t})^2}{2} \right) dt \\ &= 1 + \frac{2}{\epsilon^2}(a + \sqrt{\pi a} + 1) \end{aligned}$$

□

Now, we prove Theorem 5. The idea is the same as before, apply regret decomposition and bound $T_i(n)$, the number of times arm i is pulled. We split it into two terms, one which corresponds to the number of times index of optimal arm is less than $\mu_1 - \varepsilon$ and the other indicating the number of times arm i was pulled when its index was greater than $\mu_1 - \varepsilon$, for some ε .

Proof. of Theorem 5

$$T_i(n) = \sum_{t=1}^n \mathbb{I}\{A_t = i\} \leq \sum_{t=1}^n \mathbb{I}\left\{\hat{\mu}_1(t-1) + \sqrt{\frac{2 \log f(t)}{T_1(t-1)}} \leq \mu_1 - \varepsilon\right\} \quad (13)$$

$$+ \sum_{t=1}^n \mathbb{I}\left\{\hat{\mu}_i(t-1) + \sqrt{\frac{2 \log f(t)}{T_i(t-1)}} \geq \mu_1 - \varepsilon \text{ and } A_t = i\right\} \quad (14)$$

We bound the expectation of the first term.

$$\mathbb{E}\left[\sum_{t=1}^n \mathbb{I}\left\{\hat{\mu}_1(t-1) + \sqrt{\frac{2 \log f(t)}{T_1(t-1)}} \leq \mu_1 - \varepsilon\right\}\right] \quad (15)$$

$$\leq \sum_{t=1}^n \sum_{s=1}^n \mathbb{P}\left(\hat{\mu}_{1s} + \sqrt{\frac{2 \log f(t)}{s}} \leq \mu_1 - \varepsilon\right) \quad (16)$$

$$\leq \sum_{t=1}^n \sum_{s=1}^n \exp\left(-\frac{s\left(\sqrt{2 \log f(t)/s} + \varepsilon\right)^2}{2}\right) \quad (17)$$

$$\leq \sum_{t=1}^n \frac{1}{f(t)} \sum_{s=1}^n \exp\left(-\frac{s\varepsilon^2}{2}\right) \leq \frac{5}{\varepsilon^2} \quad (18)$$

Now on to the second term.

$$\mathbb{E}\left[\sum_{t=1}^n \mathbb{I}\left\{\hat{\mu}_i(t-1) + \sqrt{\frac{2 \log \mathbf{f}(t)}{T_i(t-1)}} \geq \mu_1 - \varepsilon \text{ and } A_t = i\right\}\right] \quad (19)$$

$$\leq \mathbb{E}\left[\sum_{t=1}^n \mathbb{I}\left\{\hat{\mu}_i(t-1) + \sqrt{\frac{2 \log \mathbf{f}(n)}{T_i(t-1)}} \geq \mu_1 - \varepsilon \text{ and } A_t = i\right\}\right] \quad (20)$$

$$\leq \mathbb{E}\left[\sum_{s=1}^n \mathbb{I}\left\{\hat{\mu}_{is} + \sqrt{\frac{2 \log f(n)}{T_i(t-1)}} \geq \mu_1 - \varepsilon\right\}\right] \quad (21)$$

$$= \mathbb{E}\left[\sum_{s=1}^n \mathbb{I}\left\{\hat{\mu}_{is} - \mu_i + \sqrt{\frac{2 \log f(n)}{T_i(t-1)}} \geq \Delta_i - \varepsilon\right\}\right] \quad (22)$$

$$\leq 1 + \frac{2(\log f(n) + \sqrt{\pi \log f(n)} + 1)}{(\Delta_i - \varepsilon)^2} \quad (23)$$

The last inequality is obtained by applying Lemma 1. Then this gives us

$$\mathbb{E}[T_i(n)] \leq 1 + \frac{5}{\varepsilon^2} + \frac{2\left(\log f(n) + \sqrt{\pi \log f(n)} + 1\right)}{(\Delta_i - \varepsilon)^2}$$

which implies Inequality 10 since ε was arbitrary. For Inequality 11, choose $\varepsilon = \log^{-1/4}(n)$ and take the limit as n tends to infinity. $\square$

6 Information Theoretic Lower Bounds

Any Bandit Algorithm which learns the optimal arm can only do so by pulling suboptimal arms as well. In a sense, we need a minimum number of pulls to gather information about the reward distributions. We formalize this idea.

Definition 4. *The worst-case regret of a policy π in an environment class $\mathcal{E}$ is defined as*

$$R_n(\pi, \mathcal{E}) = \sup_{\nu \in \mathcal{E}} R_n(\pi, \nu)$$

To encapsulate the notion of hardness of the environment class itself, we define the following.

Definition 5. *The Minimax Regret of an environment class $\mathcal{E}$ is defined as*

$$R_n^*(\mathcal{E}) = \inf_{\pi \in \Pi} \sup_{\nu \in \mathcal{E}} R_n(\pi, \nu)$$

This means that no matter how optimal a policy is, there is a bandit instance in this environment for which the policy suffers at least this much regret.

Consider the following example in the environment class $\mathcal{E}_G^k(1)$ where the reward distributions are Gaussians with mean $\mu_a \in [0, 1]$ and variance 1. Assume n is sufficiently larger than k . We want to prove the known bound on minimax regret, i.e. $R_n^*(\mathcal{E}_G^k(1)) \geq c\sqrt{kn}$ for a universal constant c . It suffices to show that for every policy π , there exist bandit instances with means $\mu, \mu' \in [0, 1]^k$ so that $\max\{R_n(\pi, \nu), R_n(\pi, \nu')\} \geq c\sqrt{kn}$.

Let $\Delta \in (0, 1/2]$, to be fixed later and choose $\mu = (\Delta, 0, \dots, 0)$. Now we want to choose μ' so that it is similar to μ , but π is not optimal for it. So, if $i = \arg \min_{a \in \mathcal{A}} \mathbb{E}[T_a(n)]$, i.e. the arm with least expected number of pulls according to π , then choose $\mu' = (\Delta, 0, \dots, 2\Delta, \dots, 0)$ with 2Δ in the i^{th} position. Observe

$$n = \mathbb{E}[T_1(n)] + \sum_{a \neq 1} \mathbb{E}[T_a(n)] \geq (k-1)\mathbb{E}[T_i(n)]$$

So choose $\Delta \geq \sqrt{\frac{k-1}{n}}$ to get the result.

Next, we prove the Bretagnole-Huber inequality that is used later.

Lemma 2. *Let P and Q be probability measures on the same measurable space. Then for any event A ,*

$$P(A) + Q(A^c) \geq \frac{1}{2} \exp(-KL(P, Q))$$

Proof. Let $\nu = P + Q$ and write $p = \frac{dP}{d\nu}$, $q = \frac{dQ}{d\nu}$ (these exist since $P, Q \ll \nu$). We prove the stronger bound

$$\int p \wedge q \geq \frac{1}{2} \exp(-D(P\|Q)),$$

which implies the statement because for any measurable A , $\int p \wedge q = \int_A p \wedge q + \int_{A^c} p \wedge q \leq \int_A p + \int_{A^c} q = P(A) + Q(A^c)$. Start with Le Cam's inequality:

$$\int p \wedge q \geq \frac{1}{2} \left(\int \sqrt{pq} \right)^2. \quad (1)$$

To prove this, observe that $pq = (p \wedge q)(p \vee q)$. Then,

$$\left(\int \sqrt{pq} \right)^2 = \left(\int \sqrt{(p \wedge q)(p \vee q)} \right)^2 \leq \left(\int p \wedge q \right) \left(\int p \vee q \right) \leq 2 \left(\int p \wedge q \right)$$

Now lower-bound the RHS of (1) by an exponential of the negative KL. Write $\left(\int \sqrt{pq} \right)^2 = \exp(2 \log \int \sqrt{pq})$ and apply Jensen's inequality to the concave map $\log$ (equivalently move $\log$ inside and use convexity of $\exp$):

$$\begin{aligned} \left(\int \sqrt{pq} \right)^2 &= \exp\left(2 \log \int \sqrt{pq}\right) \\ &\geq \exp\left(2 \int p \log \sqrt{\frac{q}{p}}\right) \\ &= \exp\left(- \int p \log \frac{p}{q}\right) \\ &= \exp(-D(P\|Q)). \end{aligned} \quad (2)$$

(The second line uses Jensen: $\log \int \sqrt{pq} \geq \int p \log \sqrt{q/p}$; the algebraic steps follow since $p > 0$ implies $q > 0$ a.e. on that set.)

Combine (1) and (2):

$$\int p \wedge q \geq \frac{1}{2} \exp(-D(P\|Q)).$$

□

6.1 Minimax Lower Bounds

Lemma 3. (*Divergence Decomposition*) Let $\nu = (P_1, \dots, P_k)$ and $\nu' = (P'_1, \dots, P'_k)$. By fixing a policy π , we obtain the probability measures $\mathbb{P}_{\nu, \pi}$ and $\mathbb{P}_{\nu', \pi}$. Then,

$$KL(\mathbb{P}_{\nu, \pi} \| \mathbb{P}_{\nu', \pi}) = \sum_{a \in \mathcal{A}} \mathbb{E}[T_a(n)] KL(P_a \| P'_a)$$

Proof. For a trajectory $(a_{1:n}, x_{1:n})$, the joint density under ν and policy π is

$$p_{\nu, \pi}(a_{1:n}, x_{1:n}) = \prod_{t=1}^n \pi_t(a_t | a_{1:t-1}, x_{1:t-1}) p_{a_t}(x_t),$$

and under ν' it is the same expression with p'_{a_t} in place of p_{a_t} . Thus,

$$\log \frac{p_{\nu, \pi}(a_{1:n}, x_{1:n})}{p_{\nu', \pi}(a_{1:n}, x_{1:n})} = \sum_{t=1}^n \log \frac{p_{a_t}(x_t)}{p'_{a_t}(x_t)}.$$

Taking expectation under $\mathbb{P}_{\nu, \pi}$ gives

$$\begin{aligned} KL(\mathbb{P}_{\nu, \pi} \| \mathbb{P}_{\nu', \pi}) &= \mathbb{E}_{\nu, \pi} \left[\sum_{t=1}^n \log \frac{p_{A_t}(X_t)}{p'_{A_t}(X_t)} \right] \\ &= \sum_{t=1}^n \mathbb{E}_{\nu, \pi} \left[\log \frac{p_{A_t}(X_t)}{p'_{A_t}(X_t)} \right]. \end{aligned}$$

Conditioning on the chosen action:

$$\begin{aligned} \mathbb{E}_{\nu, \pi} \left[\log \frac{p_{A_t}(X_t)}{p'_{A_t}(X_t)} \right] &= \sum_{a=1}^k \mathbb{E}_{\nu, \pi} \left[\log \frac{p_a(X_t)}{p'_a(X_t)} \mid A_t = a \right] \mathbb{P}_{\nu, \pi}(A_t = a) \\ &= \sum_{a=1}^k KL(P_a \| P'_a) \mathbb{P}_{\nu, \pi}(A_t = a). \end{aligned}$$

Summing over t ,

$$\begin{aligned} KL(\mathbb{P}_{\nu, \pi} \| \mathbb{P}_{\nu', \pi}) &= \sum_{t=1}^n \sum_{a=1}^k KL(P_a \| P'_a) \mathbb{P}_{\nu, \pi}(A_t = a) \\ &= \sum_{a=1}^k KL(P_a \| P'_a) \sum_{t=1}^n \mathbb{P}_{\nu, \pi}(A_t = a). \end{aligned}$$

Finally, $\sum_{t=1}^n \mathbb{P}_{\nu, \pi}(A_t = a) = \mathbb{E}_{\nu, \pi}[T_a(n)]$, yielding

$$KL(\mathbb{P}_{\nu, \pi} \| \mathbb{P}_{\nu', \pi}) = \sum_{a=1}^k \mathbb{E}_{\nu, \pi}[T_a(n)] KL(P_a \| P'_a).$$

□

Now we apply these techniques to actually formalize the Minimax regret on the class of Gaussian bandits, that we motivated earlier.

Theorem 6. Let $k > 1$ and $n \geq k - 1$. Then for any policy π on the environment class $\mathcal{E}_G^k(1)$, there exists $\mu \in [0, 1]^k$ such that

$$R_n(\pi, \nu_\mu) \geq \frac{1}{27} \sqrt{(k-1)n}$$

Proof. Following the construction outlined in the example at the start of the section, we select two environments ν_μ and $\nu_{\mu'}$ as follows: let arm 1 be optimal with gap $\Delta \in (0, \frac{1}{2}]$, and let $i \in \{2, \dots, k\}$ be the arm with smallest expected number of pulls among all suboptimal arms under π . Then define ν_μ so that arm 1 has mean Δ and every other arm has mean 0, and define $\nu_{\mu'}$ to be identical except that arm i has mean 2Δ . Then we know that,

$$\mathbb{E}_{\nu_\mu, \pi}[T_i(n)] \leq \frac{n}{k-1}.$$

Only arm i differs across the two environments, so by the divergence decomposition and the KL divergence formula for Gaussians with unit variance,

$$\begin{aligned}\text{KL}(\mathbb{P}_{\nu_\mu, \pi} \parallel \mathbb{P}_{\nu_{\mu'}, \pi}) &= \mathbb{E}_{\nu_\mu, \pi}[T_i(n)] \text{KL}(\mathcal{N}(0, 1) \parallel \mathcal{N}(2\Delta, 1)) \\ &= \mathbb{E}_{\nu_\mu, \pi}[T_i(n)] \frac{(2\Delta)^2}{2} \\ &= 2\Delta^2 \mathbb{E}_{\nu_\mu, \pi}[T_i(n)] \\ &\leq 2\Delta^2 \frac{n}{k-1}.\end{aligned}$$

Let

$$A = \{T_i(n) \leq n/2\}.$$

Applying the Bretagnolle-Huber inequality,

$$\mathbb{P}_{\nu_\mu, \pi}(A) + \mathbb{P}_{\nu_{\mu'}, \pi}(A^c) \geq \frac{1}{2} \exp\left(-\text{KL}(\mathbb{P}_{\nu_\mu, \pi} \parallel \mathbb{P}_{\nu_{\mu'}, \pi})\right).$$

Under ν_μ , arm i has gap Δ , so whenever A^c holds (i.e. $T_i(n) > n/2$),

$$\begin{aligned}R_n(\pi, \nu_\mu) &\geq \Delta \mathbb{E}_{\nu_\mu, \pi}[\mathbf{1}_{A^c} T_i(n)] \\ &\geq \Delta \cdot \frac{n}{2} \mathbb{P}_{\nu_\mu, \pi}(A^c).\end{aligned}$$

Under $\nu_{\mu'}$, arm 1 has gap Δ , so on A ,

$$\begin{aligned}R_n(\pi, \nu_{\mu'}) &\geq \Delta \mathbb{E}_{\nu_{\mu'}, \pi}[\mathbf{1}_A T_1(n)] \\ &\geq \Delta \cdot \frac{n}{2} \mathbb{P}_{\nu_{\mu'}, \pi}(A).\end{aligned}$$

Thus,

$$\begin{aligned}\max\{R_n(\pi, \nu_\mu), R_n(\pi, \nu_{\mu'})\} &\geq \frac{1}{2}(R_n(\pi, \nu_\mu) + R_n(\pi, \nu_{\mu'})) \\ &\geq \frac{\Delta n}{4}(\mathbb{P}_{\nu_{\mu'}, \pi}(A) + \mathbb{P}_{\nu_\mu, \pi}(A^c)).\end{aligned}$$

Applying Bretagnolle-Huber and the KL bound,

$$\max\{R_n(\pi, \nu_\mu), R_n(\pi, \nu_{\mu'})\} \geq \frac{\Delta n}{8} \exp\left(-2\Delta^2 \frac{n}{k-1}\right).$$

We now maximize the lower bound in Δ . Define

$$g(\Delta) = \Delta n \exp\left(-2\Delta^2 \frac{n}{k-1}\right).$$

Let $x = \Delta \sqrt{\frac{n}{k-1}}$, then

$$g(\Delta) = \sqrt{(k-1)n} x e^{-2x^2}.$$

The maximizer is $x = \frac{1}{2\sqrt{2}}$, and

$$x e^{-2x^2} = \frac{1}{2\sqrt{2}} e^{-1/4} \geq \frac{1}{27}.$$

Therefore,

$$\max\{R_n(\pi, \nu_\mu), R_n(\pi, \nu_{\mu'})\} \geq \frac{1}{27} \sqrt{(k-1)n}.$$

Finally, the maximizing Δ is

$$\Delta = \frac{1}{2\sqrt{2}} \sqrt{\frac{k-1}{n}} \leq \frac{1}{2},$$

so it defines a valid $\mu \in [0, 1]^k$. This completes the proof. $\square$

6.2 Instance Dependence Lower Bounds

Definition 6. A policy π is called consistent over a class of bandits E if $\forall \nu \in E$ and $p > 0$, t holds that

$$\lim_{n \rightarrow \infty} \frac{R_n(\pi, \nu)}{n^p}$$

A class of consistent policies over $\mathcal{E}$ is denoted $\Pi_{\text{cons}}(\mathcal{E})$

For each suboptimal arm i , define

$$d_i = \inf \{ D(P'_i \| P_i) : P'_i \in M_i, \mu(P'_i) > \mu^* \}.$$

This quantity represents the smallest possible Kullback–Leibler divergence between the true distribution of arm i and any alternative distribution under which arm i becomes optimal. Intuitively, d_i measures how difficult it is to distinguish the true instance from one in which arm i is the best arm. Any consistent policy must pull arm i often enough to reliably separate these two cases; otherwise it cannot remain consistent on the alternative instance. The Lai–Robbins lower bound formalizes this requirement in terms of d_i .

Theorem 7. Let $\mathcal{E} = \mathcal{M}_1 \times \dots \times \mathcal{M}_k$ and $\pi \in \Pi_{\text{cons}}(\mathcal{E})$ be a consistent policy over E . Then, $\forall \nu = (P_i)_{i=1}^k \in \mathcal{E}$, it holds that,

$$\liminf_{n \rightarrow \infty} \frac{R_n(\pi, \nu)}{\log(n)} \geq \sum_{i: \Delta_i > 0} \frac{\Delta_i}{d_{\text{inf}}(P_i, \mu^*, \mathcal{M}_i)}$$

Proof. Fix a suboptimal arm i and $\varepsilon > 0$. By the definition of d_i there exists $P'_i \in M_i$ with

$$D(P'_i \| P_i) \leq d_i + \varepsilon \quad \text{and} \quad \mu(P'_i) > \mu^*.$$

Let $\nu' = (P_1, \dots, P_{i-1}, P'_i, P_{i+1}, \dots, P_k)$; under ν' arm i is strictly optimal.

For $n \geq 1$ and $\delta \in (0, 1)$ define the event

$$A_n = \left\{ T_i(n) \leq \frac{(1 - \delta) \log n}{d_i + \varepsilon} \right\}.$$

By the divergence decomposition (Lemma 4) and the choice of P'_i ,

$$\begin{aligned} \text{KL}(\mathbb{P}_{\nu, \pi} \| \mathbb{P}_{\nu', \pi}) &= \mathbb{E}_{\nu, \pi}[T_i(n)] D(P_i \| P'_i) \\ &\leq \mathbb{E}_{\nu, \pi}[T_i(n)] (d_i + \varepsilon). \end{aligned}$$

Apply the Bretagnolle–Huber inequality to $\mathbb{P}_{\nu, \pi}$, $\mathbb{P}_{\nu', \pi}$ and the event A_n :

$$\begin{aligned} \mathbb{P}_{\nu, \pi}(A_n) + \mathbb{P}_{\nu', \pi}(A_n^c) &\geq \frac{1}{2} \exp \left(- \text{KL}(\mathbb{P}_{\nu, \pi} \| \mathbb{P}_{\nu', \pi}) \right) \\ &\geq \frac{1}{2} \exp \left(- (d_i + \varepsilon) \mathbb{E}_{\nu, \pi}[T_i(n)] \right). \end{aligned}$$

Assume for contradiction that

$$\liminf_{n \rightarrow \infty} \frac{\mathbb{E}_{\nu, \pi}[T_i(n)]}{\log n} < \frac{1}{d_i + \varepsilon}.$$

Then there exists $\delta \in (0, 1)$ and an infinite sequence of n for which

$$\mathbb{E}_{\nu, \pi}[T_i(n)] \leq \frac{(1 - \delta) \log n}{d_i + \varepsilon}. \tag{24}$$

For such n the KL bound above yields

$$\text{KL}(\mathbb{P}_{\nu, \pi} \| \mathbb{P}_{\nu', \pi}) \leq (d_i + \varepsilon) \mathbb{E}_{\nu, \pi}[T_i(n)] \leq (1 - \delta) \log n,$$

and so Bretagnolle–Huber implies

$$\mathbb{P}_{\nu, \pi}(A_n) + \mathbb{P}_{\nu', \pi}(A_n^c) \geq \frac{1}{2} n^{-(1 - \delta)}.$$

Consistency of π on ν' (and the fact that arm i is strictly optimal under ν') implies that for any fixed constant $C > 0$

$$\mathbb{P}_{\nu', \pi}(T_i(n) \leq C \log n) \rightarrow 0,$$

hence in particular $\mathbb{P}_{\nu',\pi}(A_n^c) \rightarrow 1$. Therefore for all sufficiently large n in the subsequence of (24),

$$\mathbb{P}_{\nu,\pi}(A_n) \geq \frac{1}{2}n^{-(1-\delta)} - (1 - \mathbb{P}_{\nu',\pi}(A_n^c)) \geq \frac{1}{4}n^{-(1-\delta)},$$

where the last inequality holds for large n because $\mathbb{P}_{\nu',\pi}(A_n^c) \rightarrow 1$.

We now derive a contradiction by relating this tail probability to the expectation. Let

$$m_n := \left\lfloor \frac{(1-\delta)\log n}{d_i + \varepsilon} \right\rfloor.$$

Using the tail-sum formula for nonnegative integer random variables,

$$\begin{aligned} \mathbb{E}_{\nu,\pi}[T_i(n)] &= \sum_{t \geq 1} \mathbb{P}_{\nu,\pi}(T_i(n) \geq t) \\ &\geq \sum_{t=1}^{m_n} \mathbb{P}_{\nu,\pi}(T_i(n) \geq m_n) = m_n \mathbb{P}_{\nu,\pi}(T_i(n) \geq m_n). \end{aligned}$$

But $\mathbb{P}_{\nu,\pi}(T_i(n) \geq m_n) = 1 - \mathbb{P}_{\nu,\pi}(A_n)$, so for those n we obtain

$$\mathbb{E}_{\nu,\pi}[T_i(n)] \geq m_n (1 - \mathbb{P}_{\nu,\pi}(A_n)) \geq m_n \left(1 - \frac{1}{4}n^{-(1-\delta)}\right).$$

Since $m_n = \Theta(\log n)$ and $m_n n^{-(1-\delta)} \rightarrow 0$, this gives for large n in the subsequence

$$\mathbb{E}_{\nu,\pi}[T_i(n)] \geq m_n - o(1).$$

But by definition $m_n \geq (1-\delta)\log n/(d_i + \varepsilon) - 1$, so along the subsequence the lower bound above contradicts the assumed upper bound (24) for all sufficiently large n . This contradiction shows that the assumption was false.

Therefore, for every $\varepsilon > 0$,

$$\liminf_{n \rightarrow \infty} \frac{\mathbb{E}_{\nu,\pi}[T_i(n)]}{\log n} \geq \frac{1}{d_i + \varepsilon}.$$

Letting $\varepsilon \downarrow 0$ yields

$$\liminf_{n \rightarrow \infty} \frac{\mathbb{E}_{\nu,\pi}[T_i(n)]}{\log n} \geq \frac{1}{d_i}.$$

Finally, summing over suboptimal arms and multiplying by the corresponding gaps gives

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{R_n(\pi, \nu)}{\log n} &= \liminf_{n \rightarrow \infty} \frac{\sum_{i: \Delta_i > 0} \Delta_i \mathbb{E}_{\nu,\pi}[T_i(n)]}{\log n} \\ &\geq \sum_{i: \Delta_i > 0} \frac{\Delta_i}{d_i}, \end{aligned}$$

which completes the proof. $\square$

Hence, we make the following definition for optimality.

Definition 7. A policy π is asymptotically optimal if

$$\lim_{n \rightarrow \infty} \frac{R_n(\nu, \pi)}{\log n} = \sum_{a: \Delta_a > 0} \frac{\Delta_a}{KL_{inf}(P_a, \mu^*)}$$

For most classes, we can specifically design policies due to the fundamental nature of the quantity d_{inf} . KL-UCB is the modified version of UCB that is asymptotically optimal for not only subgaussian distributions, but also exponential families.

7 Pure Exploration: Best-Arm Identification

We now change goals: instead of minimizing cumulative regret, we want to *identify* the best arm.

Definition 8. An algorithm is (ε, δ) -PAC if it returns an arm $\hat{a}$ satisfying $\mathbb{P}(\mu^* - \mu_{\hat{a}} \leq \varepsilon) \geq 1 - \delta$.

Two settings:

- Fixed-Confidence: target confidence δ , minimize samples.
- Fixed-Budget: fixed number of samples, minimize failure probability.

We'll concentrate on the first one.

7.1 Naive uniform-sampling

Sample each arm $\ell = O(\epsilon^{-2} \log(k/\delta))$ times; return empirical best. Sample complexity: $O(k\epsilon^{-2} \log(k/\delta))$.

Proof. Let a be an arm such that $\mu_a < \mu^* - \epsilon$. We want to bound $\mathbb{P}(\hat{\mu}_a > \hat{\mu}_{a^*})$

$$\begin{aligned} \mathbb{P}(\hat{\mu}_a > \hat{\mu}_{a^*}) &\leq \mathbb{P}(\hat{\mu}_a > \mu_a + \epsilon/2 \text{ OR } \hat{\mu}_{a^*} < \mu^* - \epsilon/2) \\ &\leq \mathbb{P}(\hat{\mu}_a > \mu_a + \epsilon/2) + \mathbb{P}(\hat{\mu}_{a^*} < \mu^* - \epsilon/2) \\ &\leq e^{-(\frac{\epsilon}{2})^2 l} \end{aligned}$$

where l is the number of samples of arm a . We have the first inequality since it can be shown that,

$$\hat{\mu}_a > \hat{\mu}_{a^*} \implies \hat{\mu}_a > \mu_a + \epsilon/2 \text{ OR } \hat{\mu}_{a^*} < \mu^* - \epsilon/2$$

And the last inequality comes from Hoeffding's inequality. Choosing $l = \frac{4}{\epsilon^2} \log(\frac{2k}{\delta})$, we get that $\mathbb{P}(\hat{\mu}_a > \hat{\mu}_{a^*}) \leq \frac{\delta}{k}$. Using union bounds, we see that the algorithm is (ϵ, δ) -PAC with sample complexity $O(\frac{k}{\epsilon^2} \log \frac{k}{\delta})$ $\square$

The problem with this algorithm is that every arm is given an equal number of chances. Instead, we would like to eliminate the poorly performing arms earlier in the algorithm, i.e. to make the algorithm sensitive to the outcomes of the samples. This leads us onto the next algorithm.

7.2 Successive Elimination

Algorithm (SuccElim): Maintain active set S . Repeat: sample each arm in S once, compute empirical means, remove arms whose empirical mean is much worse than the current best by a confidence-dependent threshold. Stop when one arm remains.

This improves on naive sampling by adaptively eliminating bad arms early.

Algorithm 4 Successive Elimination(δ)

- 1: Set $t = 1$ and $S = A$
 - 2: Set for every arm a : $\hat{\mu}_a^1 = 0$.
 - 3: **repeat**
 - 4: Sample every arm $a \in S$ once and let $\hat{\mu}_a^t$ be the average reward of arm a by time t .
 - 5: Let $\hat{\mu}_{max}^t = \max_{a \in S} \hat{\mu}_a^t$ and $\alpha_t = \sqrt{\frac{\log(cnt^2/\delta)}{t}}$
 - 6: **for** every arm $a \in S$ such that $\hat{\mu}_{max}^t - \hat{\mu}_a^t \geq 2\alpha_t$ **do**
 - 7: set $S = S \setminus a$
 - 8: **end for**
 - 9: $t = t + 1$.
 - 10: **until** $|S| > 1$
 - 11: **output** S .
-

Theorem 8. *The Successive Elimination algorithm is a $(0, \delta)$ -PAC algorithm, and with probability at least $1 - \delta$ its arm sample complexity is bounded by $O\left(\sum_{i=2}^n \frac{\log(\frac{n}{\delta \Delta_i})}{\Delta_i^2}\right)$.*

Proof. For any time t and action $a \in S_t$ we have that,

$$\mathbb{P}[|\hat{\mu}_a^t - \mu_a| \geq \alpha_t] \leq e^{-\alpha_t^2 t} \leq \frac{\delta}{cnt^2}.$$

where $c = \sum_{t=1}^{\infty} \frac{1}{t^2} = \pi^2/6$.

Therefore, by applying union bounds, the best arm is never eliminated with a probability of $1 - \delta$. Furthermore, since α_t approaches zero as t increases, eventually every non-best arm is eliminated. Thus the algorithm is $(0, \delta)$ -PAC. It remains to compute the arm sample complexity. The definition of α_t combined with the assumption that $|\hat{\mu}_a^t - \mu_a| \leq \alpha_t$ yields that

$$\Delta_i - 2\alpha_t = (\mu_1 - \alpha_t) - (\mu_i + \alpha_t) \geq \hat{\mu}_1 - \hat{\mu}_i \geq 2\alpha_t,$$

which holds with probability at least $1 - \frac{\delta}{n}$ for

$$t_i = O\left(\frac{\log(n/\delta \Delta_i)}{\Delta_i^2}\right).$$

The arm sample complexity is $t_2 + \sum_{i=2}^n t_i$, which completes the proof. $\square$

Remark 8. We can modify the successive elimination algorithm so that it is (ϵ, δ) -PAC. Instead of stopping when only one arm survives the elimination, we stop when either only one arm remains or when each of the k surviving arms were sampled $O(\frac{1}{\epsilon^2} \log(\frac{k}{\delta}))$. In the latter case the algorithm returns the best arm so far. In this case we can show that the algorithm finds an ϵ -optimal arm with probability at least $1 - \delta$ after

$$O\left(\sum_{i: \Delta_i > \epsilon} \frac{\log(\frac{n}{\delta \Delta_i})}{\Delta_i^2} + \frac{N(\Delta, \epsilon)}{\epsilon^2} \log\left(\frac{N(\Delta, \epsilon)}{\delta}\right)\right),$$

where $N(\Delta, \epsilon) = |\{i \mid \Delta_i < \epsilon\}|$ is the number of arms which are ϵ -optimal.

7.3 Median Elimination

Median Elimination is a phase-based algorithm achieving optimal *uniform* sample complexity up to constants.

Algorithm (MedianElim): Iterate phases. In phase ℓ with active set S_ℓ , sample each arm enough to estimate means to accuracy ϵ_ℓ , compute empirical medians, discard the half with smaller empirical means, and tighten parameters until one arm remains.

Algorithm 5 Median Elimination(ϵ, δ)

```

1: Set  $S = A$ .  $\epsilon_1 = \epsilon/4$ ,  $\delta_1 = \delta/2$ ,  $\ell = 1$ .
2: repeat
3:   Sample every arm  $a \in S$  for  $\frac{1}{(\epsilon_\ell/2)^2} \ln(3/\delta_\ell)$  times, and let  $\hat{\mu}_a^\ell$  denote its empirical value.
4:   Find the median of  $\hat{\mu}_a^\ell$ , denoted by  $m_\ell$ .
5:    $S_{\ell+1} = S_\ell \setminus \{a : \hat{\mu}_a^\ell < m_\ell\}$ .
6:   if  $|S_\ell| = 1$  then
7:     output  $S_\ell$ 
8:   else
9:      $\epsilon_{\ell+1} = \frac{3}{4}\epsilon_\ell$ ;  $\delta_{\ell+1} = \delta_\ell/2$ ;  $\ell = \ell + 1$ ;
10:    Go to 3.
11:  end if
```

Theorem 9. The Median Elimination(ϵ, δ) algorithm is an (ϵ, δ) -PAC algorithm and its arm sample complexity is $O(\frac{n \ln(\frac{1}{\delta})}{\epsilon^2})$.

First we show that in the ℓ -th phase the expected reward of the best arm in S_ℓ decreases by at most ϵ_ℓ .

Lemma 4. For the Median Elimination(ϵ, δ) algorithm we have that

$$\mathbb{P}[\max_{j \in S_\ell} \mu_j \leq \max_{i \in S_{\ell+1}} \mu_i + \epsilon_\ell] \geq 1 - \delta_\ell$$

Proof. WLOG, we look at the first round and assume that μ_1 is the reward of the best arm. We bound the failure probability using the event $E_1 = \{\hat{\mu}_1 < \mu_1 - \epsilon_1/2\}$. By step 3 of the algorithm, we sample sufficiently such that $\mathbb{P}[E_1] \leq \delta_1/3$.

In case E_1 does not hold,

$$\mathbb{P}[\hat{\mu}_j \geq \hat{\mu}_1 \mid \hat{\mu}_1 \geq \mu_1 - \epsilon_1/2] \leq \mathbb{P}[\hat{\mu}_j \geq \mu_j + \epsilon_1/2 \mid \hat{\mu}_1 \geq \mu_1 - \epsilon_1/2] \leq \delta_1/3$$

Let $\#bad$ be the number of arms which are not ϵ_1 -optimal but are empirically better than the best arm. We have that $E[\#bad \mid \hat{\mu}_1 \geq \mu_1 - \epsilon_1/2] \leq n\delta_1/3$. Next we apply Markov inequality to obtain,

$$\mathbb{P}[\#bad \geq n/2 \mid \hat{\mu}_1 \geq \mu_1 - \epsilon_1/2] \leq \frac{n\delta_1/3}{n/2} = 2\delta_1/3.$$

Using the union bound gives us that the probability of failure is bounded by δ_1 . □

Next we prove that arm sample complexity is bounded by $O(\frac{n \ln(\frac{1}{\delta})}{\epsilon^2})$.

Lemma 5. The Median Elimination(ϵ, δ) has $O(\frac{n \ln(\frac{1}{\delta})}{\epsilon^2})$ arm sample complexity.

Proof. The number of arm samples in the ℓ -th round is $\frac{4n_\ell \ln(3/\delta_\ell)}{\epsilon_\ell^2}$. By definition we have that $\delta_\ell = \delta/2^\ell, n_\ell = n/2^{\ell-1}, \epsilon_\ell = (\frac{3}{4})^{\ell-1} \epsilon/4$. Therefore we have

$$\begin{aligned} \sum_{\ell=1}^{\log_2(n)} \frac{n_\ell \ln(3/\delta_\ell)}{(\epsilon_\ell/2)^2} &= 4 \sum_{\ell=1}^{\log_2(n)} \frac{n/2^{\ell-1} \ln(2^\ell 3/\delta)}{((\frac{3}{4})^{\ell-1} \epsilon/4)^2} \\ &= 64 \sum_{\ell=1}^{\log_2(n)} n \left(\frac{8}{9}\right)^{\ell-1} \left(\frac{\ln(1/\delta)}{\epsilon^2} + \frac{\ln(3)}{\epsilon^2} + \frac{\ell \ln(2)}{\epsilon^2}\right) \\ &\leq 64 \frac{n \ln(1/\delta)}{\epsilon^2} \sum_{\ell=1}^{\infty} \left(\frac{8}{9}\right)^{\ell-1} (\ell C' + C) \leq O\left(\frac{n \ln(1/\delta)}{\epsilon^2}\right) \end{aligned}$$

□

Proof. {of theorem 9} From Lemma 5 we have that the sample complexity is bounded by $O(\frac{n \ln(1/\delta)}{\epsilon^2})$. By Lemma 4 we have that the algorithm fails with probability δ_i in each round so that over all rounds the probability of failure is bounded by $\sum_{i=1}^{\log_2(n)} \delta_i \leq \delta$. In each round we reduce the optimal reward of the surviving arms by at most ϵ_i so that the total error is bounded by $\sum_{i=1}^{\log_2(n)} \epsilon_i \leq \epsilon$. □

7.4 Exponential-Gap Elimination (EGE)

EGE is an instance-adaptive algorithm that groups arms by gap scales and eliminates many arms quickly. Key ideas:

- Work in rounds indexed by r ; target gap scale 2^{-r} .
- Run MedianElim as a subroutine to find a leader in the current set.
- Eliminate arms whose empirical gap exceeds a threshold relative to the leader.

Algorithm 6 Exponential-Gap Elimination

Require: confidence $\delta > 0$

- 1: initialize $S_1 \leftarrow [n], r \leftarrow 1$
 - 2: **while** $|S_r| > 1$ **do**
 - 3: let $\varepsilon_r = 2^{-r}/4$ and $\delta_r = \delta/(50r^3)$
 - 4: sample each arm $i \in S_r$ for $t_r = (2/\varepsilon_r^2) \ln(2/\delta_r)$ times, and let $\hat{\mu}_i^r$ be the average reward
 - 5: invoke $i_r \leftarrow \text{Median Elimination}(S_r, \varepsilon_r/2, \delta_r)$ and let $\hat{\mu}_*^r = \hat{\mu}_{i_r}^r$
 - 6: set $S_{r+1} \leftarrow S_r \setminus \{i \in S_r : \hat{\mu}_i^r < \hat{\mu}_*^r - \varepsilon_r\}$
 - 7: update $r \leftarrow r + 1$
 - 8: **end while**
 - 9: **return** arm in $S_r = 0$
-

Essentially, the only modification required is limiting the number of rounds performed by the algorithm to $O(\log(1/\varepsilon))$.

Theorem 10. *With probability at least $1 - \delta$, Algorithm 1 identifies the optimal arm using*

$$O\left(\sum_{i=2}^n \frac{1}{\Delta_i^2} \log\left(\frac{1}{\delta} \log \frac{1}{\Delta_i}\right)\right)$$

arm pulls.

We note that a simple modification³ of Algorithm 1 give rise to an (ε, δ) -PAC algorithm (i.e., algorithm that finds an ε -optimal arm with probability $\geq 1 - \delta$) with similar, almost-optimal guarantees.

We first establish a few lemmas.

Lemma 6. *With probability at least $1 - \delta/5$, we have $\hat{\mu}_1^r \geq \hat{\mu}_*^r - \varepsilon_r$ for all r .*

This can be proven using union bounds and Hoeffding's inequality in a very similar manner to the other proofs seen so far.

For the rest of the analysis we will need some additional notation. For all $0 \leq s \leq \lceil \log_2(1/\Delta) \rceil$, we define the set

$$A_s = \{i \in [n] : 2^{-s} \leq \Delta_i < 2^{-s+1}\}$$

and let $n_s = |A_s|$. Also, we denote the set of arms from A_s surviving after round r by $S_{r,s} = S_r \cap A_s$, for all $r, s \geq 0$. Next, we prove that from round s onwards, a constant fraction of the surviving arms of the set A_s is eliminated on each round.

Lemma 7. Assume that the optimal arm is not eliminated by the algorithm. Then with probability at least $1 - 4\delta/5$, we have $|S_{r,s}| \leq \frac{1}{8}|S_{r-1,s}|$ for all $1 \leq s \leq r$.

Proof. Consider round r . The guarantees of the Median Elimination algorithm imply that $\mu_{i_r} \geq \mu_1 - \varepsilon_r/2$ with probability at least $1 - \delta_r$, assuming that the best arm reached round r . We have

$$\mathbb{P}[\hat{\mu}_*^r \leq \mu_1 - \varepsilon_r] \leq 2\delta_r.$$

For any arm i with $\Delta_i \geq 2^{-r} = 4\varepsilon_r$, the event $\hat{\mu}_i^r \geq \hat{\mu}_*^r - \varepsilon_r$ implies that either $\hat{\mu}_i^r \geq \mu_i + \varepsilon_r$ or $\hat{\mu}_*^r \leq \mu_1 - \varepsilon_r$. Hence,

$$\mathbb{P}[\hat{\mu}_i^r \geq \hat{\mu}_*^r - \varepsilon_r] \leq \mathbb{P}[\hat{\mu}_i^r \geq \mu_i + \varepsilon_r] + \mathbb{P}[\hat{\mu}_*^r \leq \mu_1 - \varepsilon_r] \leq 3\delta_r$$

which means that an arm $i \in A_s$ survives round $r \geq s$ with probability at most $3\delta_r$. Consequently, we have

$$\mathbb{E}[|S_{r,s}|] \leq 3\delta_r |S_{r-1,s}|,$$

The lemma now follows via a union bound, by which the probability of failure is bounded by $\sum_{r=1}^{\infty} \sum_{s=1}^r 24\delta_r = \sum_{r=1}^{\infty} 24\delta/(50r^2) < 4\delta/5$. $\square$

We next calculate how many times an arm from A_s is pulled by the algorithm, ignoring (for now) the pulls spent due to invocations of Median Elimination.

Lemma 8. With probability at least $1 - \delta$, the total number of times an arm from A_s is sampled in line 4 is $O(4^s n_s \log(s/\delta))$ for all s .

Proof. Let T_s denote the total number of times an arm from A_s is pulled. If the algorithm is successful we have

$$T_s = \sum_{r=1}^{\infty} |S_{r,s}| \cdot t_r \leq \sum_{r=1}^{s-1} |A_s| \cdot t_r + \sum_{r=s}^{\infty} |S_{r,s}| \cdot t_r \leq n_s \sum_{r=1}^{s-1} t_r + n_s \sum_{r=0}^{\infty} \left(\frac{1}{8}\right)^{r+1} t_{r+s}.$$

Now, the first sum is upper bounded by

$$\sum_{r=1}^{s-1} t_r = 32 \sum_{r=1}^{s-1} 4^r \ln \frac{100r^3}{\delta} = O\left(4^s \log \frac{s}{\delta}\right).$$

For the second sum, we have

$$\begin{aligned} \sum_{r=0}^{\infty} \left(\frac{1}{8}\right)^{r+1} t_{r+s} &= 4^{s+1} \sum_{r=0}^{\infty} 2^{-r} \ln \frac{100(r+s)^3}{\delta} \\ &\leq 3 \cdot 4^{s+1} \ln \frac{4s}{\delta} \sum_{r=0}^{\infty} 2^{-r} + 3 \cdot 4^{s+1} \sum_{r=1}^{\infty} 2^{-r} \ln(5r) \\ &= O\left(4^s \log \frac{s}{\delta}\right). \end{aligned}$$

$\square$

Now, for the theorem.

Proof. We first prove that the algorithm is correct with probability $1 - \delta$: The lemmas imply that any suboptimal arm is eliminated eventually and ensures that the best arm is never eliminated. Hence, the algorithm terminates at some point and returns the correct arm. By a union bound, this happens with probability at least $1 - \delta$.

Now for the sample complexity of the algorithm: The number of pulls spent by the invocation of Median Elimination on round r is $O(|S_{r-1}| \cdot t_r)$, which is the same as the total number of pulls used by the algorithm itself (in line 4) on that round. Hence, the invocations of Median Elimination affect the overall sample complexity only by a constant factor.

Now, if the algorithm is successful, the number of times it pulls an arm from A_s is $T_s = O(4^s n_s \log(s/\delta))$ for all s . Recalling that $2^s < 2/\Delta_i$ for all $i \in A_s$, we obtain

$$T_s = O\left(4^s n_s \log \frac{s}{\delta}\right) = O\left(\sum_{i \in A_s} \frac{1}{\Delta_i^2} \log \left(\frac{1}{\delta} \log \frac{1}{\Delta_i}\right)\right)$$

and the theorem follows by summing over s . $\square$

References

- [1] Eyal Even-Dar, Shie Mannor, and Yishay Mansour. Pac bounds for multi-armed bandit and markov decision processes. *The Annals of Mathematics and Artificial Intelligence*, 28(1):55–81, 2006.
- [2] Aurélien Garivier and Emilie Kaufmann. Optimal best arm identification with fixed confidence. In *Proceedings of the 29th Conference on Learning Theory (COLT)*, pages 998–1027. PMLR, 2016.
- [3] Zohar Karnin, Tomer Koren, and Oren Somekh. Almost optimal exploration in multi-armed bandits. In *Proceedings of the 30th International Conference on Machine Learning (ICML)*, volume 28, pages 1238–1246. PMLR, 2013.
- [4] Tor Lattimore and Csaba Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.